Problem 4 — NearHelp

Overview

When someone has a sudden medical emergency, a car breakdown, a gas leak, or needs urgent help — they call one person. But the fastest help is almost always a neighbour they have never spoken to. Build a platform that activates your immediate community in a crisis.

Required Features & Score Distribution

#	Feature	Marks (out of 20)
1	**SOS Broadcasting** — Trigger SOS with crisis type and auto-detected location. Broadcasts via WebSockets to users within a configurable radius (500 m, 1 km, 2 km). Pulsing pin appears live on the map.	5/20
2	**Responder Flow** — Nearby users tap *"I'm Responding"*; broadcaster sees live responder locations. Per-responder chat for coordination. Broadcaster marks SOS resolved — pin disappears in real time.	4/20
3	**AI Crisis Assistant** — AI surfaces first-response guidance based on crisis type. Generates a pre-filled emergency summary ready to read to services. Post-resolution debrief prompt.	4/20
4	**Skill Registry & Community Layer** — Users register skills (CPR, doctor, nurse, etc.); skilled responders highlighted during active SOS. Community resource map (AEDs, fire extinguishers). Nearby emergency services auto-surfaced on SOS trigger.	3/20
5	**Trust & Verification** — Responders build a rated response history. False alerts flagged and penalised; repeat offenders suspended. Anonymous mode for threat-based SOS.	2/20
6	**Admin Dashboard** — Live city-wide SOS view. Response-time analytics by locality. Misuse flagging and suspension management.	2/20

Optional Features

- **Offline SMS Fallback** — SOS triggered via SMS when internet is unavailable; posts to platform on user's behalf.
- **Guardian Mode** — Guardians assigned to vulnerable users are notified before the wider community.
- **Post-Crisis Welfare Check** — Automated check-in message sent to broadcaster 24 hours after resolution.

Technical Expectations

- Backend: real-time server with WebSocket support (Socket.io, `ws`, etc.) and a persistent database.
- Frontend: map integration (Leaflet, Mapbox, or Google Maps) with live marker updates.
- AI Integration: any LLM API (OpenAI, Anthropic, Gemini, etc.) for the Crisis Assistant.
- Geospatial queries must be efficient — use indexed radius queries or a spatial database where appropriate.
- Authenticated endpoints must enforce user identity; anonymous SOS must not expose PII to responders.